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FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

on

"Smart Finance Manager & Personal Expense Intelligence System"

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CERTIFICATE

This is to certify that the project based laboratory report entitled "**Smart Finance Manager & Personal Expense Intelligence System**" submitted by **Sheri Abhishek reddy, K.Yashwanth , Gelam Sai, Poorna sai nallapaneni** bearing Regd. No. **2500031276, 2500031321, 2500031409, 2500031453** to the **Department of COMPUTER SCIENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in **Python Programming (25SC2007)** course in I B Tech II Semester, is a Bonafede record of the work carried out by them under my supervision during the academic year 2025-2026.

PROJECT SUPERVISOR

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ABSTRACT

The **"Smart Finance Manager & Personal Expense Intelligence System (SFM-PEIS)"** is a Python-based application developed to help users efficiently manage and analyze their daily financial activities. The system allows users to record expenses, categorize spending, and generate meaningful insights for better financial planning.

This project integrates multiple Python programming concepts such as string processing, lists, dictionaries, functions, object-oriented programming, file handling using CSV and JSON, exception handling, and modular programming. It also includes a rule-based recommendation system to provide intelligent suggestions based on user spending behavior.

Over a span of ten weeks, the project was developed step by step, where each week focuses on implementing a specific concept and integrating it into the overall system. The system supports data cleaning, storage, analysis, and reporting, making it a comprehensive personal finance management tool.

Key Words: Finance, Expenses, Saving, Spending

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INTRODUCTION

Welcome to the "**Smart Finance Manager & Personal Expense Intelligence System (SFM-PEIS)"**, a simple and efficient application developed using Python to manage personal financial activities. This system is designed to help users track their daily expenses, analyze spending patterns, and make better financial decisions. The application provides a user-friendly interface where users can enter their income and record expenses under different categories. It allows users to view summaries such as total expenditure, average spending, and most frequently used expense categories. The system also supports reading data from CSV files and storing data securely using JSON format. In addition, the program performs expense analysis and provides intelligent recommendations based on the user's spending habits. Graphical representation of data is also included to help users understand their financial behavior visually. This project integrates multiple programming concepts such as object-oriented programming, file handling, data processing, and visualization, making it a comprehensive and practical financial management tool.

Features of Python used in this project:

- Easy to learn and use
- Supports structured and modular programming
- Produces efficient programs
- Provides powerful libraries like matplotlib
- Platform independent and portable

AIM

The main purpose of the **Smart Finance Manager & Personal Expense Intelligence System (SFM-PEIS)** is to provide an efficient way for users to manage and track their personal expenses. It is an automated system that helps in recording daily financial transactions and analyzing spending patterns. The system reduces manual effort by storing data digitally and performing automatic calculations such as total expense, average spending, and category-wise analysis. It also provides insights and recommendations to help users maintain a balanced financial lifestyle. One of the key features of the system is the ability to store and retrieve data using files (CSV and JSON), ensuring that user information is safely maintained and can be accessed anytime.

MODULES IN THE PROJECT

The aims of this project are:

- To provide an easy way to record and manage expenses anytime
- To reduce manual calculation of financial data
- To analyze spending patterns and categorize expenses
- To store and retrieve data using file handling techniques
- To provide graphical representation of expenses
- To give intelligent recommendations based on user spending
- To improve personal financial planning and savings

ADVANTAGES

- Reduces manual work in managing financial records
- Eliminates the use of paper by storing data digitally
- Provides accurate and error-free calculations
- Allows easy storage and retrieval of data using files
- Provides graphical visualization for better understanding

DISADVANTAGES

- Requires basic knowledge of computer usage
- Data may be lost if files are deleted accidentally
- Limited to local storage (no cloud access)
- Graph feature requires additional library (matplotlib)

FUTURE IMPLEMENTATION

- Integration with mobile or web application
- Automatic monthly expense alerts and notifications
- Cloud storage for secure data backup
- Advanced data visualization and reports
- AI-based financial suggestions for better savings

SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS:

- The major software requirements of the project are as follows:

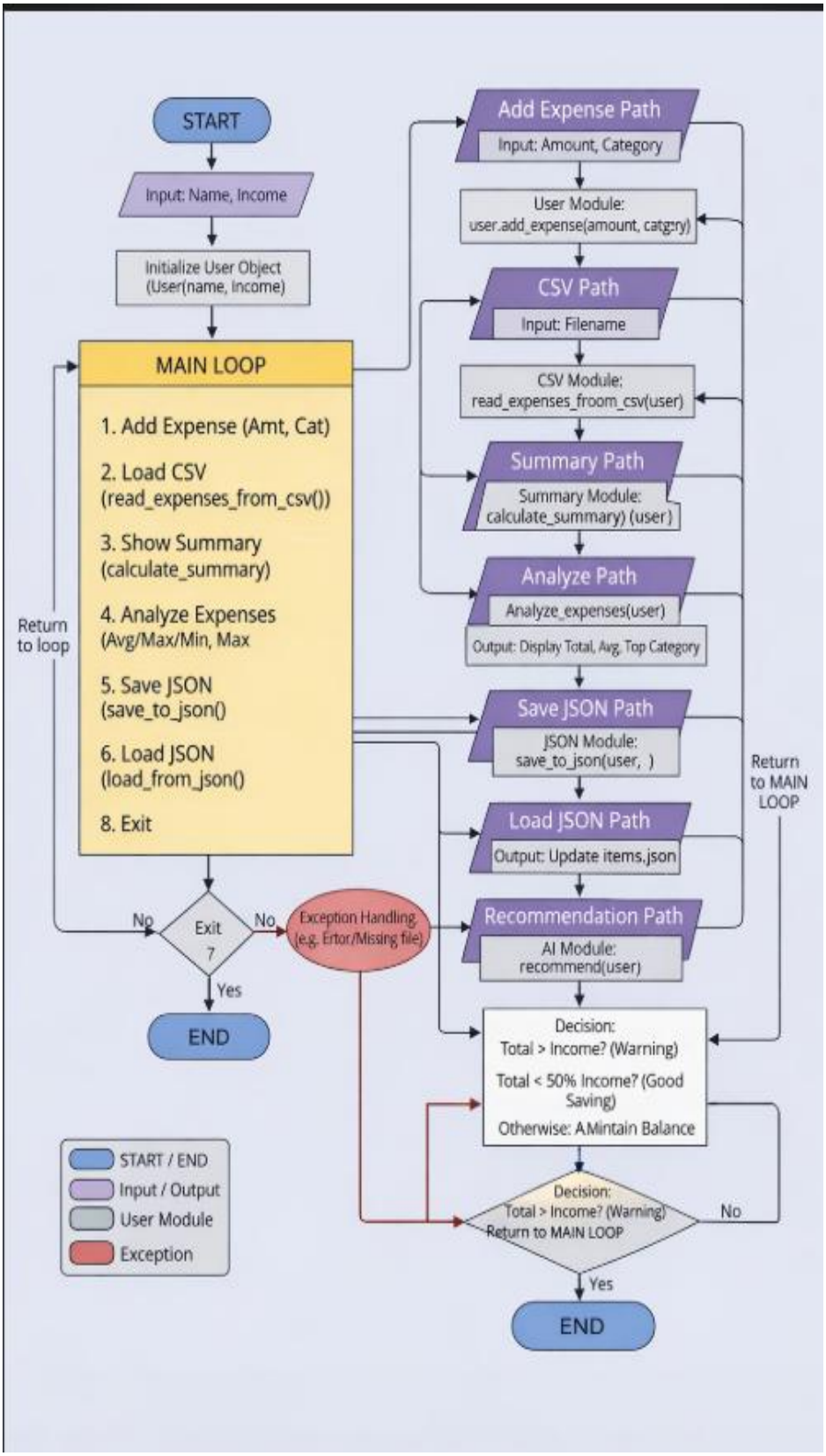
- **Programming Language:** Python
- **Operating System:** Windows 10 or above / Linux / MacOS
- **Libraries Used:** matplotlib, csv, json
- **Development Tool:** Any Python IDE (IDLE / VS Code / PyCharm)

HARDWARE REQUIREMENTS:

- The hardware requirements for running the system are as follows:

- **RAM:** Minimum 4 GB
- **Processor:** Intel i3 or above
- **System Type:** 64-bit Computer
- **Input Devices:** Keyboard, Mouse
- **Output Device:** Monitor

DATA FLOW DIAGRAM



ALGORITHM

Step-1: Start

Step-2: Read user name and income

Step-3: Create user object

Step-4: Display menu options:

1. Add Expense
2. Load from CSV
3. Show Summary
4. Analyze Expenses
5. Save to JSON
6. Load from JSON
7. Recommendation
8. Show Graph
9. Exit

Step-5: Read user choice

Step-6: If choice = 1,

Read expense amount and category

Add expense to user

Step-7: If choice = 2,

Read CSV file and load data

Step-8: If choice = 3,

Calculate total, average, and category frequency

Display summary

Step-9: If choice = 4,

Analyze expenses (min, max, average)

Step-10: If choice = 5,

Save data to JSON file

Step-11: If choice = 6,

Load data from JSON file

Step-12: If choice = 7,

Provide recommendation based on spending

Step-13: If choice = 8,

Display graph of expenses

Step-14: If choice = 9,

Stop program

IMPLEMENTATION

```
import csv
import json
import matplotlib.pyplot as plt
class User:
    def __init__(self, name, income):
        self.name = name
        self.income = income
        self.expenses = []
    def add_expense(self, amount, category):
        self.expenses.append((amount, category))

def calculate_summary(user):
    total = sum([e[0] for e in user.expenses])
    avg = total / len(user.expenses) if user.expenses else 0
    freq = {}
    for _, cat in user.expenses:
        freq[cat] = freq.get(cat, 0) + 1
    most_used = max(freq, key=freq.get) if freq else "None"
    return total, avg, freq, most_used

def clean_text(text):
    stop_words = ["on", "the", "a", "is"]
    clean = ""
    for ch in text:
        if ch.isalnum() or ch.isspace():
            clean += ch.lower()
    words = clean.split()
    filtered = [w for w in words if w not in stop_words]
    return " ".join(filtered)
def read_expenses_from_csv(filename, user):
    try:
```

```
        with open(filename, "r") as file:
            reader = csv.reader(file)
            for row in reader:
                try:
                    amount = float(row[0])
                    category = row[1]
                    user.add_expense(amount, category)
                except:
                    print("Invalid row:", row)
    except FileNotFoundError:
        print("CSV file not found")

def save_to_json(user, filename):
    data = {
        "name": user.name,
        "income": user.income,
        "expenses": user.expenses
    }
    with open(filename, "w") as f:
        json.dump(data, f)

def load_from_json(filename):
    try:
        with open(filename, "r") as f:
            data = json.load(f)
            user = User(data["name"], data["income"])
            user.expenses = data["expenses"]
            return user
    except:
        print("Error loading data")
        return None
-
def remove_duplicates(expenses):
    unique = []
```

```
    for e in expenses:
        if e not in unique:
            unique.append(e)
    return unique

def recommend(user):
    total, _, _, _ = calculate_summary(user)
    if total > user.income:
        print("Warning: You are overspending!")
    elif total < user.income * 0.5:
        print("Good saving habit!")
    else:
        print("Maintain better balance")

def analyze_expenses(user):
    amounts = [e[0] for e in user.expenses]
    if not amounts:
        print("No expenses found")
        return
    print("Average Expense:", sum(amounts)/len(amounts))
    print("Max Expense:", max(amounts))
    print("Min Expense:", min(amounts))
    for i, amt in enumerate(amounts):
        if amt > 500:
            print("Day", i+1, ": High Spending")
        elif amt >= 200:
            print("Day", i+1, ": Normal")
        else:
            print("Day", i+1, ": Low")

def show_graph(user):
    categories = []
    amounts = []
    for amt, cat in user.expenses:
```

```
        categories.append(cat)
        amounts.append(amt)
    if not categories:
        print("No data to show graph")
        return
    plt.bar(categories, amounts, color='blue')
    plt.xlabel("Categories")
    plt.ylabel("Amount")
    plt.title("Expense Analysis Graph")
    plt.show()

def main():
    name = input("Enter your name: ")
    income = float(input("Enter your income: "))
    user = User(name, income)
    while True:
        print("\n1.Add Expense")
        print("2.Load from CSV")
        print("3.Show Summary")
        print("4.Analyze Expenses")
        print("5.Save to JSON")
        print("6.Load from JSON")
        print("7.Recommendation")
        print("8.Show Graph")
        print("9.Exit")
        ch = int(input("Enter choice: "))
        if ch == 1:
            amt = float(input("Enter amount: "))
            cat = input("Enter category: ")
            user.add_expense(amt, cat)
        elif ch == 2:
            file = input("Enter CSV file name: ")
            read_expenses_from_csv(file, user)
        elif ch == 3:
```

```
        total, avg, freq, most = calculate_summary(user)
        print("Total:", total)
        print("Average:", avg)
        print("Category Frequency:", freq)
        print("Most Used Category:", most)
    elif ch == 4:
        analyze_expenses(user)
    elif ch == 5:
        save_to_json(user, "finance.json")
        print("Saved successfully")
    elif ch == 6:
        user = load_from_json("finance.json")
        print("Data loaded")
    elif ch == 7:
        recommend(user)
    elif ch == 8:
        show_graph(user)
    elif ch == 9:
        break
main()
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

Adding expense

```
C:\Users\hp\PyCharmMiscProject\.ven
Enter your name: Ravi
Enter your income: 5000

1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 1
Enter amount: 200
Enter category: food
```

Adding expense

```
1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 1
Enter amount: 100
Enter category: bills
```

Adding expense

```
1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 1
Enter amount: 600
Enter category: travel
```

Summary

```
1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 3
Total: 900.0
Average: 300.0
Category Frequency: {'food': 1, 'travel': 1, 'bills': 1}
Most Used Category: food
```

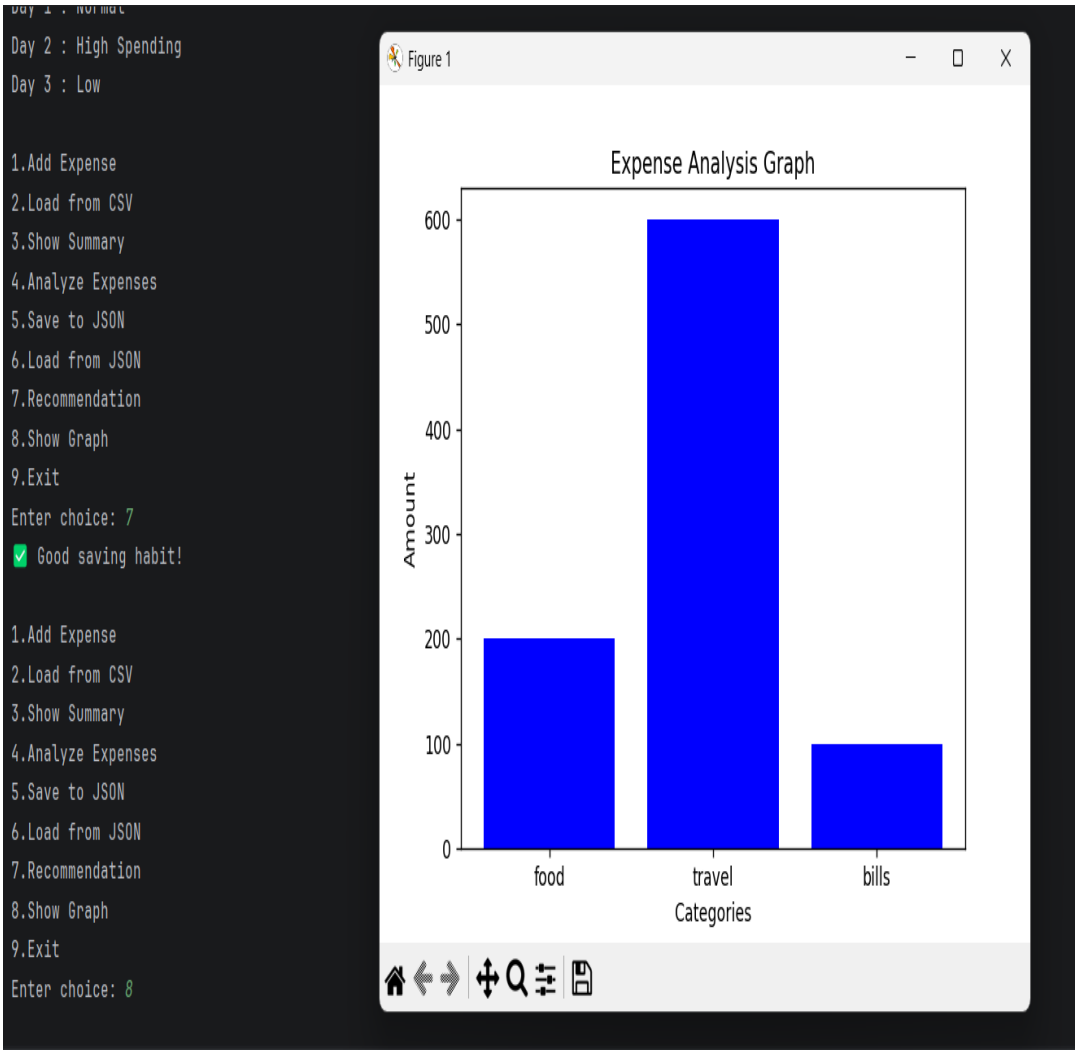
Analyzing expenses

```
1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 4
Average Expense: 300.0
Max Expense: 600.0
Min Expense: 100.0
Day 1 : Normal
Day 2 : High Spending
Day 3 : Low
```

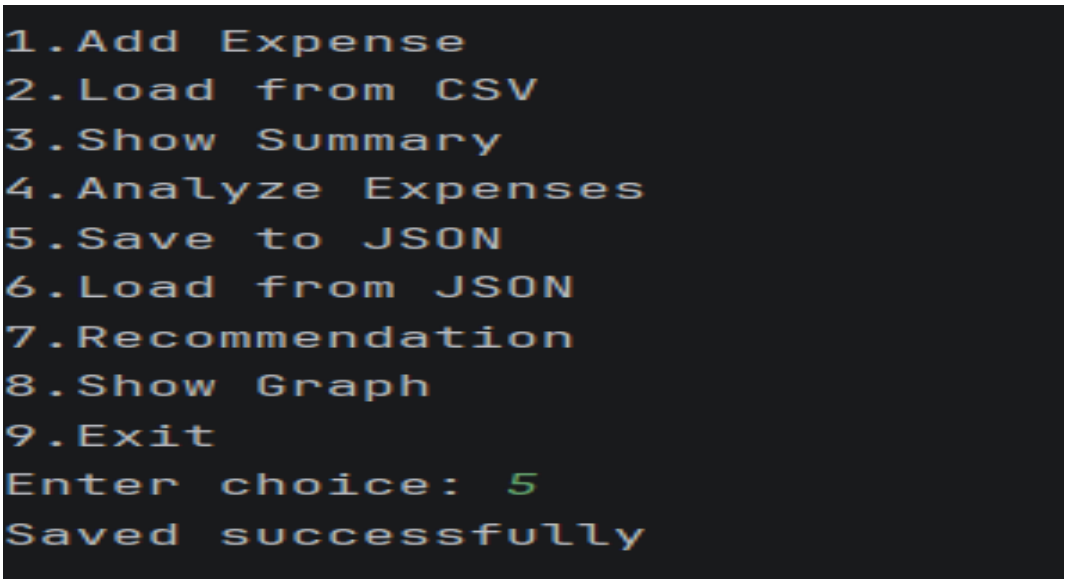
Recommendation

```
1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 7
✅ Good saving habit!
```

Graph representation



Saving to json



Json file

```
_init_.py    project.py    expenses.csv    finance.json

1  {"name": "Ravi", "income": 5000.0, "expenses": [[200.0, "food"], [600.0, "travel"], [100.0, "bills"]]}
```

Exit

```
1.Add Expense
2.Load from CSV
3.Show Summary
4.Analyze Expenses
5.Save to JSON
6.Load from JSON
7.Recommendation
8.Show Graph
9.Exit
Enter choice: 9

Process finished with exit code 0
```

CONCLUSION

This project has been developed successfully and the performance of the system has been found satisfactory. The "**Smart Finance Manager & Personal Expense Intelligence System"** helps users to easily manage and track their daily expenses without much effort.

The system provides quick access to financial information such as total expenses, average spending, and category-wise analysis, saving the user's time and improving accuracy. The user-friendly interface allows easy interaction for adding, viewing, and analyzing expenses. Overall, this project demonstrates how Python can be effectively used to build a simple and efficient personal finance management system, helping users make better financial decisions.